

Electric Vehicle-Integrated Optimal Power Flow for Emergency Charging Coordination and Power System Resilience

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Abstract—Large-scale electric vehicle (EV) charging during emergency evacuations can place considerable stress on power systems when demand surges are compounded by degraded network conditions. Existing optimal power flow (OPF) formulations typically neglect the flexibility and priority of EV charging under emergency conditions. This paper proposes an EV-aware DC OPF framework in which aggregated EV charging is modeled as a prioritized controllable load. A priority-based curtailment mechanism is integrated into the formulation to enable selective reduction of non-critical charging while preserving system feasibility and ensuring service to critical demand. The proposed approach is evaluated on the IEEE RTS 24-bus system under evacuation-driven demand and contingency conditions. Results demonstrate that coordinated EV charging eliminates involuntary load shedding, maintains system feasibility under stressed operating conditions, and preserves high-priority charging services by leveraging demand flexibility within the OPF framework. These improvements enhance power system resilience under emergency conditions.

Index Terms—Electric vehicles, emergency charging, optimal power flow, power system resilience, priority-based load control

I. INTRODUCTION

The increasing penetration of electric vehicles (EVs) is fundamentally changing the interaction between power systems and transportation infrastructure [1]. Unlike traditional loads, EVs introduce highly flexible yet potentially synchronized electricity demand that is closely tied to human mobility patterns [2,3]. During extreme events such as wildfires, hurricanes, and floods, large-scale evacuations can trigger abrupt and geographically concentrated EV charging demand precisely when power system infrastructure is degraded or partially unavailable [4].

Recent studies have demonstrated that wildfire-induced EV evacuation behavior can significantly affect power system resilience. Power system resilience is defined as the ability of the grid to withstand disturbances, maintain operational feasibility, and recover rapidly from disruptions while minimizing service interruptions [5-7]. Within this context, detailed modeling of EV evacuation during wildfires shows that both pre-departure and post-arrival charging demand can induce substantial spatial and temporal shifts in transmission system loading, increasing congestion and operational stress [8].

These results indicate that EV charging during emergencies represents a distinct operational challenge that cannot be adequately represented by conventional demand models.

To address the strong coupling between transportation and power systems, several works have proposed integrated optimization frameworks that jointly model traffic flow, charging behavior, and power system operation. In this context, dynamic traffic assignment combined with stochastic or robust unit commitment formulations has been used to coordinate evacuation routing and generation scheduling under uncertainty [9,10]. In parallel, prior studies have investigated EV-integrated power system planning considering distributed generation, energy storage systems, and charging infrastructure through multi-objective optimization frameworks [11]. While these approaches provide comprehensive system-level insights, their computational complexity limits their applicability to planning or offline analysis, making them unsuitable for real-time operation. Moreover, EV charging demand is often represented in a simplified manner, without explicitly modeling its flexibility and priority-dependent characteristics under contingency conditions.

Furthermore, resilience-oriented power system studies have focused on assessing and mitigating infrastructure vulnerability under extreme events. Probabilistic wildfire modeling and spatiotemporal fragility analysis have been developed to evaluate transmission system resilience and identify critical corridors [5]. Other works have explored resilience enhancement strategies based on mobile energy resources, mobile generators, and vehicle-to-grid (V2G) technologies using stochastic or multi-stage optimization frameworks [7,12]. At the distribution level, EV charging station placement and planning models have been proposed to improve resilience and post-disaster restoration capabilities [13].

Despite these advances, a critical operational gap remains. Most existing studies focus on planning, probabilistic assessment, or multi-stage optimization, whereas system operators require fast, transparent, and computationally efficient tools for real-time decision-making during emergencies. In practice, optimal power flow (OPF) models are widely used to manage congestion and minimize load shedding under abnormal condi-

tions. However, conventional OPF formulations typically treat load as fixed or uniformly curtailable, limiting their ability to represent the flexible and prioritized nature of EV charging demand during emergency scenarios.

To address this gap, an EV-aware DC optimal power flow (DCOPF) framework is developed in this paper for the operational coordination of emergency EV charging. In the proposed formulation, aggregated EV charging stations are modeled as prioritized, dispatchable loads, enabling selective curtailment of non-critical charging demand while preserving system feasibility during contingency conditions. The framework is evaluated on the IEEE RTS 24-bus transmission system under evacuation-induced EV demand spikes, varying EV penetration levels, and N-1 contingencies, demonstrating that coordinated EV-aware operation can prevent infeasible system states, eliminate involuntary load shedding, and maintain essential EV charging services. The resulting optimization problem is formulated as a convex quadratic program (QP), which can be solved efficiently using standard optimization solvers and is therefore efficient due to its convex QP structure.

The main contributions of this paper are summarized as follows:

- An EV-aware DCOPF formulation that models emergency EV charging demand as a prioritized, dispatchable load within power system operation.
- A priority-based curtailment mechanism that enables selective coordination of EV charging demand under contingency conditions.
- A comprehensive evaluation under evacuation-driven demand and a representative N-1 contingency scenario based on a critical transmission line outage, demonstrating improved system feasibility, reduced involuntary load shedding, and enhanced EV service continuity.

II. SYSTEM MODEL AND EV CHARGING REPRESENTATION

This section describes the transmission network model and the representation of EV charging demand considered in this study. The IEEE 24-bus Reliability Test System (RTS) is used as the base network, adopted from the MATPOWER library [14]. The corresponding data file (`case24_ieee_rts`) is publicly available in the MATPOWER repository. The test system is modified to include aggregated EV charging stations at selected buses. The modeling framework is designed to capture emergency-driven charging demand and its impact on system operation under contingency conditions.

A. Power System Model

The study is conducted on the IEEE 24-bus RTS, which is a standard benchmark to evaluate transmission-level power system operation. The network includes buses, generators, and transmission lines with predefined operational limits, enabling a realistic assessment of congestion and load balancing under stressed conditions. The system data are obtained from MATPOWER [14]. Generator cost functions are modeled as

TABLE I
EV CHARGING MODEL PARAMETERS

Parameter	Value
Number of EV buses	8
EV bus selection	Buses with highest load demand
Normal demand factor	$\alpha_{normal} = 0.1$
Spike demand factor	$\alpha_{evacuation} = 0.4$
High-priority weight	$w_{high} = 5$
Medium-priority weight	$w_{medium} = 2$
Low-priority weight	$w_{low} = 1$

quadratic functions, and transmission lines are subject to thermal flow limits.

A DCOPF formulation is adopted to capture system-wide active power balance while maintaining computational efficiency during emergency and contingency conditions. In this approximation, voltage magnitudes are assumed to be constant and reactive power effects are neglected, resulting in a linear relationship between power flows and voltage phase angles. This modeling approach is widely used for operational studies and contingency analysis in transmission systems due to its computational efficiency and scalability [14,15].

Network contingencies are represented through a representative N-1 contingency scenario, in which a critical transmission line is removed from service. The selected contingency corresponds to the most heavily loaded line under stressed operating conditions, enabling the evaluation of system response to a realistic and impactful outage.

B. EV Charging Station Representation

Emergency EV charging demand is represented through aggregated charging stations connected to a subset of buses in the IEEE RTS-24 system. Specifically, 8 buses are selected from the highest-demand load buses to represent major charging locations under evacuation conditions. This selection reflects the spatial concentration of EV charging demand in areas with high population and infrastructure density.

For each selected bus i , the maximum EV charging demand is modeled as a fraction of the local base load as follows:

$$\bar{P}_i^{EV} = \alpha P_i^{\text{load}} \quad (1)$$

where $\alpha = \alpha_{normal} = 0.1$ and $\alpha = \alpha_{evacuation} = 0.4$ represent the scaling factors for EV charging demand under normal conditions and evacuation scenarios, respectively. This formulation enables scalable representation of EV demand while preserving tractability in the DCOPF framework. The key parameters of the EV charging model are summarized in Table I.

To capture differentiated charging urgency, EV charging stations are categorized into three priority levels: high, medium, and low. These correspond to critical services (e.g., emergency response and hospitals), general residential/commercial demand, and flexible or deferrable charging, respectively.

For a charging station located at bus i , the served EV charging power is treated as a controllable variable in the coordinated charging scenario and is constrained as follows:

$$0 \leq P_i^{EV} \leq \bar{P}_i^{EV}, \quad \forall i \in \mathcal{N}_{EV}, \quad (2)$$

where P_i^{EV} denotes the EV charging power supplied at bus i and $\bar{P}_i^{EV}$ represents the maximum available EV charging demand under the considered operating condition.

In contrast, in the uncoordinated charging scenario, EV demand is modeled as inflexible and is treated as a fixed load, as follows:

$$P_i^{EV} = \bar{P}_i^{EV}, \quad \forall i \in \mathcal{N}_{EV}. \quad (3)$$

The parameters used in this study are representative values selected to demonstrate the proposed framework and can be adjusted to reflect different EV penetration levels and operational conditions.

III. EV-AWARE DC OPTIMAL POWER FLOW FORMULATION

This section presents the mathematical formulation of the proposed EV-aware DCOPF model for coordinating EV charging under contingency conditions. In this formulation, aggregated EV charging demand is modeled as a controllable and prioritized decision variable within the network-constrained optimization problem, enabling selective curtailment to maintain system feasibility.

A. Decision Variables

The decision variables of the proposed EV-aware DCOPF are as follows:

- P_g : active power output of generator g ,
- θ_i : voltage phase angle at bus i ,
- f_l : active power flow on transmission line l ,
- P_i^{EV} : served aggregated EV charging power at bus i ,
- P_i^{LS} : involuntary load shedding at bus i .

The sets of buses, generators, and transmission lines are denoted $\mathcal{B}$, $\mathcal{G}$, and $\mathcal{L}$, respectively, and $\mathcal{E} \subseteq \mathcal{B}$ denotes the subset of buses hosting aggregated EV charging stations. Unlike conventional DCOPF, which subsumes all demand into a single fixed vector, the proposed formulation exposes P_i^{EV} as an independent decision variable. This structural choice is the operational foundation of the priority-based curtailment mechanism: reducing P_i^{EV} at a congested bus directly relaxes the local power balance without affecting the committed non-EV base load P_i^D .

B. Generation Cost Function

Generator fuel costs are modeled as quadratic functions of active power output as follows:

$$C_g(P_g) = c_{2,g} P_g^2 + c_{1,g} P_g + c_{0,g}, \quad \forall g \in \mathcal{G}, \quad (4)$$

where $c_{2,g}$, $c_{1,g}$, and $c_{0,g}$ are the quadratic, linear, and constant cost coefficients of generator g , respectively, obtained from the IEEE RTS 24-bus dataset via MATPOWER [14]. This

quadratic structure preserves the convexity of the overall optimization problem.

C. Objective Function

The objective function is formulated to minimize the weighted sum of generation cost, involuntary load shedding, and priority-weighted EV curtailment, as follows:

$$\min \sum_{g \in \mathcal{G}} C_g(P_g) + \lambda \sum_{i \in \mathcal{B}} P_i^{LS} + \mu \sum_{i \in \mathcal{E}} w_i (\bar{P}_i^{EV} - P_i^{EV}), \quad (5)$$

where λ is the load shedding penalty, μ is the EV curtailment penalty, w_i is the priority weight at EV bus i , and $\bar{P}_i^{EV} - P_i^{EV}$ is the curtailed portion of EV demand at bus i .

The penalty coefficients are chosen such that $\lambda \gg \mu$, enforcing a strict two-level hierarchy: the optimizer exhausts all available EV curtailment before resorting to involuntary load shedding. In this study, $\lambda = 1000$ \$/MW and $\mu = 100$ \$/MW, yielding a ratio of 10 : 1. The priority of EV charging demand is incorporated into the OPF objective through weighting factors $w_i \in \{w_{high}, w_{medium}, w_{low}\}$ assigned to high-, medium-, and low-priority EV buses, respectively. These weights are applied to the EV curtailment term, imposing higher penalties on curtailment at critical buses. This formulation enables resilience-oriented operation by biasing the optimization toward shedding deferrable low-priority demand while preserving high-priority charging services under constrained conditions.

D. Constraints

1) *Power Balance*: Nodal active power balance is enforced at every bus $i \in \mathcal{B}$:

$$\sum_{g \in \mathcal{G}_i} P_g - P_i^D - P_i^{EV} + P_i^{LS} = \sum_{j \in \mathcal{B}} B_{ij}(\theta_i - \theta_j), \quad (6)$$

where P_i^D is the non-EV base demand, $\mathcal{G}_i$ is the set of generators at bus i , and B_{ij} is the (i, j) entry of the DC susceptance matrix $\mathbf{B}$, constructed as follows:

$$B_{ij} = \begin{cases} \sum_{l \in \mathcal{L}_i} b_l, & i = j, \\ -b_l, & l = (i, j) \in \mathcal{L}, \\ 0, & \text{otherwise,} \end{cases} \quad (7)$$

where $b_l = 1/x_l$ is the susceptance of line l and $\mathcal{L}_i$ is the set of lines incident to bus i . The explicit separation of P_i^{EV} from P_i^D in (6) is a key feature of the proposed formulation, as it allows the optimizer to reduce P_i^{EV} when a transmission constraint becomes active, which relieves the power injection requirement at that bus without shedding committed load. The load shedding term P_i^{LS} remains available at every bus as a last-resort mechanism but is rendered costly by λ in (5).

2) *DC Power Flow*: Active power flow on line $l = (i, j)$ is modeled using the DC approximation:

$$f_l = b_l(\theta_i - \theta_j), \quad \forall l = (i, j) \in \mathcal{L}, \quad (8)$$

where voltage magnitudes are assumed uniform at 1.0 p.u. and reactive power effects are neglected. A reference bus angle is fixed to ensure a unique solution:

$$\theta_{\text{ref}} = 0. \quad (9)$$

This linear power flow model preserves the convexity of the formulation and is standard for operational studies and contingency analysis in transmission systems [14,15].

3) *Generator and Transmission Limits*: Physical operating bounds are imposed on all generators and transmission lines:

$$P_g^{\min} \leq P_g \leq P_g^{\max}, \quad \forall g \in \mathcal{G}, \quad (10)$$

$$-f_l^{\max} \leq f_l \leq f_l^{\max}, \quad \forall l \in \mathcal{L}, \quad (11)$$

where $f_l^{\max}$ is the thermal MVA rating of line l , obtained from the IEEE RTS 24-bus dataset. Network contingencies are represented through a representative N-1 contingency scenario by defining a reduced line set as follows:

$$\mathcal{L}^{\text{out}} = \mathcal{L} \setminus \{l^*\}, \quad (12)$$

where l^* is the outaged line corresponding to the most heavily loaded transmission line under stressed operating conditions. Constraints (7)–(10) are enforced over $\mathcal{L}^{\text{out}}$, reflecting the modified network topology. The removal of l^* redistributes power flows onto parallel paths, which can intensify congestion and is precisely the stress condition under which the EV coordination mechanism is most critical [16].

4) *EV Charging and Load Shedding Limits*: The feasible operating ranges of EV charging and load shedding variables are bounded as follows:

$$0 \leq P_i^{\text{EV}} \leq \bar{P}_i^{\text{EV}}, \quad \forall i \in \mathcal{E}, \quad (13)$$

$$0 \leq P_i^{\text{LS}} \leq P_i^{\text{D}}, \quad \forall i \in \mathcal{B}. \quad (14)$$

The upper bound in (13) is the scenario-dependent maximum aggregated charging demand defined in (1), i.e., $\bar{P}_i^{\text{EV}} = \alpha P_i^{\text{D}}$ under normal conditions and $\bar{P}_i^{\text{EV}} = \beta P_i^{\text{D}}$ during an evacuation spike. The lower bound of zero prevents reverse power injection. Together with the priority-weighted penalty in (5), constraint (13) allows the optimizer to continuously vary P_i^{EV} over $[0, \bar{P}_i^{\text{EV}}]$, with the degree of curtailment at each bus determined jointly by the binding network constraints and the priority weight w_i . Buses with $w_i = 5$ (high priority) are effectively protected: the optimizer incurs a large marginal penalty before curtailing them. Constraint (14) bounds involuntary shedding to the available non-EV demand, preventing over-shedding and ensuring physical consistency.

Constraints (6)–(14) together with objective (5) form a convex QP, owing to the quadratic cost functions (4) and the linearity of all constraints. The resulting convex QP is solved for each operating condition considered in the case study. For the outage scenario, the modified network topology is represented by substituting the reduced line set $\mathcal{L}^{\text{out}}$ into (8) and (11). The core operational mechanism is the interplay between the EV flexibility bounds (13) and the transmission limits (11): when a line flow constraint binds, the optimizer curtails P_i^{EV} at buses

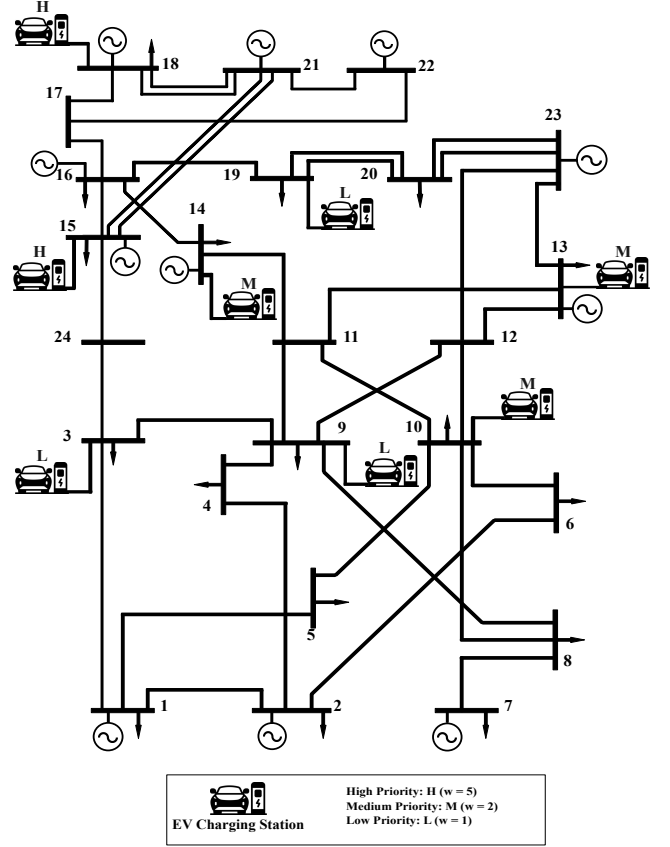


Fig. 1. IEEE RTS-24 system with EV charging stations placed at selected buses and classified into high-, medium-, and low-priority categories for coordinated EV-aware DCOPF.

contributing to the congested corridor, restoring feasibility while respecting the priority hierarchy encoded in (5).

IV. CASE STUDY AND SIMULATION RESULTS

The proposed EV-aware DCOPF framework is evaluated on the IEEE RTS-24 transmission system to assess its ability to coordinate emergency EV charging under stressed operating conditions. Aggregated EV charging stations are integrated at selected buses, as illustrated in Fig. 1, with charging demand levels varied to emulate normal and evacuation-driven conditions. The performance of the proposed coordinated framework is compared against a conventional uncoordinated DCOPF formulation in which EV charging is treated as inflexible demand. The analysis considers both normal operation and a representative N-1 contingency to evaluate system feasibility, load shedding behavior, and EV service continuity.

A. Case Definitions

Three operating conditions are considered:

- Normal operation ($\alpha_{\text{normal}} = 0.1$): Baseline system performance with nominal EV charging demand.

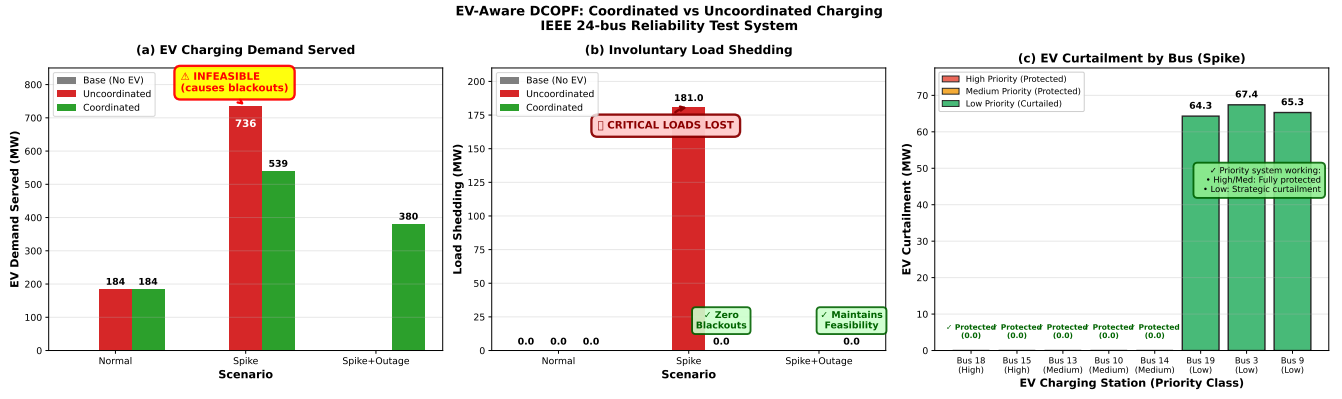


Fig. 2. EV-aware DCOPF performance on the IEEE RTS-24 system under evacuation-driven demand (a) EV charging demand served under normal operation, evacuation-induced spike, and spike with N-1 contingency (b) Involuntary load shedding comparison for uncoordinated and EV-aware operation (c) Priority-based EV charging curtailment across charging locations during evacuation spike.

- Evacuation spike ($\alpha_{evacuation} = 0.4$): Increased EV demand at selected buses to represent evacuation-driven charging.
- Spike + outage: Evacuation-driven EV demand combined with a representative N-1 contingency corresponding to the outage of a critical transmission line.

For each operating condition, two EV charging strategies are compared. In Case 1, EV charging is modeled as uncoordinated and inflexible demand. In Case 2, EV charging is treated as a controllable and prioritized demand using the proposed EV-aware DCOPF formulation. The contingency scenario corresponds to the outage of the most heavily loaded transmission line identified under stressed coordinated operation (Bus 7–Bus 8 in this case).

B. Simulation Results and Discussion

The simulation results show that uncoordinated EV charging can significantly degrade power system performance under emergency-driven demand conditions. Under normal loading, both coordinated and uncoordinated strategies serve the same EV demand without requiring load shedding, indicating that the system has sufficient flexibility to accommodate nominal charging demand. However, under evacuation-driven demand spikes, the operational difference between the two approaches becomes substantial.

As shown in Fig. 2(a), the uncoordinated strategy serves the full EV demand during the spike, but at the cost of severe transmission stress. Since EV charging is treated as fixed demand, the optimizer cannot relieve congestion through demand adjustment. Consequently, Fig. 2(b) shows that 181 MW of involuntary load shedding occurs, and the problem becomes infeasible under the combined spike and outage condition, as transmission constraints become binding.

In contrast, the proposed EV-aware DCOPF maintains system feasibility by selectively curtailing EV demand. During the spike, it serves 539 MW of EV load with zero load shedding, and under the spike-plus-outage condition, it remains feasible while serving 380 MW without shedding base load. This

demonstrates that controllable EV charging can act as an effective flexibility resource.

Fig. 2(c) shows that high- and medium-priority charging are fully preserved, while curtailment is concentrated at low-priority buses. Table II summarizes the quantitative results. Under the evacuation spike, the uncoordinated case serves 736 MW but incurs 181 MW of load shedding and reaches 100% line loading, whereas the coordinated strategy eliminates load shedding while serving 539 MW. Under the spike+outage condition, the coordinated approach remains feasible and serves 380 MW without shedding base load.

This highlights the role of EV charging as a controllable flexibility resource for maintaining feasible operation under emergency conditions.

C. Sensitivity Analysis

To further examine the resilience benefits of coordinated charging, a sensitivity analysis is performed under increasing EV demand scaling factors (α), as illustrated in Fig. 3. In particular, Fig. 3(a) shows that involuntary load shedding increases rapidly under uncoordinated charging as EV demand rises beyond the nominal spike level. By contrast, the coordinated EV-aware formulation maintains zero load shedding over the full tested range by progressively curtailing EV demand as needed.

Fig. 3(b) further shows how the served EV power is allocated across priority classes under coordinated operation. High-priority charging remains protected over the full demand range, while medium-priority demand is largely preserved and low-priority demand absorbs most of the reduction. This confirms that the proposed priority-based control mechanism remains effective across a wide range of operating conditions, including progressively increasing system stress. Overall, the case study results demonstrate that modeling EV charging as a controllable and prioritized load can substantially improve emergency power system operation. Compared with uncoordinated charging, the proposed EV-aware DCOPF restores

TABLE II
EV-AWARE DCOPF RESULTS FOR NORMAL AND EVACUATION SCENARIOS (IEEE RTS 24-Bus)

Scenario	EV Served (MW)	EV Curtailed (MW)	Load Shed (MW)	Max Line (%)	Gen Cost (\$)
Base (No EV)	0	0	0	73.1	61,001
Uncoordinated (Normal)	184	0	0	68.0	70,212
Coordinated (Normal)	184	0	0	68.0	70,212
Uncoordinated (Spike)	736	0	181	100.0	91,018
Coordinated (Spike)	539	197	0	100.0	88,938
Coordinated (Spike + Outage)	380	356	0	58.3	82,071

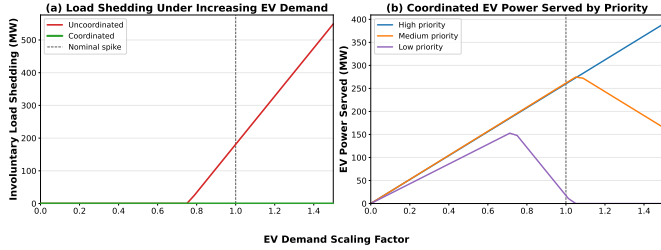


Fig. 3. Sensitivity analysis of power system resilience under varying EV demand scaling. (a) Involuntary load shedding for uncoordinated and coordinated charging. (b) Served EV power by priority under coordinated charging.

feasibility, avoids involuntary load shedding, and preserves essential EV charging services under both evacuation-driven demand spikes and contingency conditions.

V. CONCLUSION

This paper addressed emergency EV charging during large-scale evacuations, where demand spikes coincide with degraded network conditions. An EV-aware DCOPF framework was developed that models EV charging as a prioritized, controllable load, enabling selective curtailment within the OPF. Results on the IEEE RTS-24 system show that uncoordinated EV charging leads to 181 MW of load shedding and infeasibility under $N-1$ contingencies. In contrast, the proposed approach eliminates load shedding, serves 539 MW during the evacuation spike, and maintains feasibility with 380 MW under the combined spike and outage condition, while preserving high- and medium-priority demand. These results demonstrate that controlled EV charging can act as an effective flexibility resource to enhance system resilience during emergencies. Future work will extend the framework to multi-period operation with stochastic EV behavior.

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